Abhimanyu Patel

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EDUCATION

Institution	Degree / Board	Score	Rank / Distinction	Year
IIIT Bhopal	B.Tech – Computer Science & Engineering	7.71(SGPA)	Pursuing	2024–28
Orchid International School	Higher Secondary (PCMB) – Class XII	87.2%	5th Rank	2024
Sagar Public School	Secondary – Class X	97.2%	Topper	2022

PROFESSIONAL QUALIFICATIONS

CFA Institute *Professional Qualification*	• **CFA Level 1 Candidate** — registered for the CFA Program; target exam date **February 2027**	2027

PROFESSIONAL EXPERIENCE

Market Research Analyst *Finlatics (DPIIT-recognized startup, incubated at AIC-NMIMS under NITI Aayog)*	• **Spearheaded** a strategic evaluation of BigBasket's transition from scheduled delivery to Q-commerce, identifying the BB Now segment as driving **~85% of revenue mix** as of late 2025 • **Modelled** the scaling potential of high-margin private labels (BB Royal, Organic) through proposed "Experience Centres" in premium urban markets • **Designed** a B2B revenue stabilisation strategy centered on long-term MOUs with corporates for pantry management and employee gifting programs • **Assessed** the financial feasibility of a full EV fleet transition, quantifying long-term fuel and maintenance savings against upfront costs and their impact on ESG-linked IPO valuation • **Developed** the "Care Moat" framework — a data-driven route optimization and delivery-partner welfare model aimed at reducing attrition • **Delivered** a comprehensive roadmap positioning BigBasket as a sustainable "Quick Life-Essentials" platform ahead of its proposed 2026 IPO; conducted under mentorship of Nisant Mohta (Founder, Finlatics); received Letter of Recommendation	Dec 2025 – Jan 2026

PROJECTS

Options Pricing & Greeks Lab *Independent Project*	• **Engineered** an interactive options analytics platform in Python and Streamlit that prices European options under the Black-Scholes-Merton model and computes Delta, Gamma, Vega, Theta and Rho • **Formulated** an implied volatility solver (Newton-Raphson with bisection fallback) and a volatility-smile view, with safeguards for deep in-the-money, out-of-the-money and sub-intrinsic prices • **Architected** Monte Carlo (antithetic variates, standard error, confidence intervals) and Cox-Ross-Rubinstein binomial engines to price Asian and American options, benchmarking convergence against Black-Scholes • **Designed** a strategy builder with payoff and P&L diagrams, breakevens and maximum profit/loss for spreads, straddles, strangles and iron condors, alongside a Nifty 50 index option preset • **Validated** the pricing logic through an automated pytest suite covering put-call parity, finite-difference Greeks and implied volatility round-trips; deployed on Streamlit Community Cloud with a GitHub Actions keep-alive	Oct 2026
Repo Rate & Loan Impact *Independent Project*	• **Engineered** and deployed a 3-tab fintech web application on Streamlit Community Cloud that estimates a borrower's loan interest rate, EMI and total interest from CIBIL score, bank, loan type, amount and tenure • **Formulated** a floating-rate pricing model (repo rate + bank spread + CIBIL risk premium), benchmarking 7 banks across 4 loan types and ranking them by total interest cost • **Architected** a repo-rate sensitivity engine (3%–9%) quantifying before/after EMI and interest, with a "same EMI" mode that solves the revised loan tenure using the logarithmic tenure formula • **Developed** an EMI calculator with yearly and monthly amortization schedules, interactive Plotly charts and CSV export, handling edge cases such as 0% interest and non-whole-year tenures • **Validated** the model with 83 automated pytest tests (EMI against known values, schedules summing to principal, zero final balance, CIBIL band boundaries) and automated app uptime via a Playwright script on a scheduled GitHub Actions workflow	Oct 2026
Currency Fair Value Dashboard *Independent Project*	• **Engineered** and deployed an open-source Currency Fair Value Dashboard in Python 3 and pandas to quantify Forex exchange rate deviations from fundamental value, enabling data-driven currency valuation analysis • **Formulated** the valuation logic on Purchasing Power Parity (PPP) macroeconomic models, benchmarking live spot rates against PPP-implied fair value to flag over- and under-valued currencies • **Architected** a live data pipeline integrating the FRED API (macroeconomic series) and yfinance (FX spot rates), with automated pandas/NumPy transformations and interactive Plotly visualizations for reproducible analysis • **Deployed** the application as a live Streamlit web app hosted on GitHub, demonstrating end-to-end ownership from quantitative modelling to full-stack release	Sep 2026

CPSE Bond Credit Risk Intelligence Platform *Independent Project*	• **Engineered** a quantitative credit risk intelligence platform tracking 37 active bond-issuing Central Public Sector Enterprises (CPSEs), enabling systematic screening of public sector debt creditworthiness • **Developed** a financial risk evaluation module integrating issuer-level project lookups to assess capital structure stability and credit quality across state-owned debt instruments • **Designed** a primary market pipeline tracker to monitor upcoming bond issuances, categorizing real-time listing status across debt and equity markets for early-stage investment screening • **Assessed** macro-level credit risk signals across the CPSE bond universe, applying structured analytical frameworks to evaluate sovereign-linked default risk and issuer liquidity dynamics	2026
Investment Portfolio Dashboard *Independent Project*	• **Engineered** an automated portfolio management dashboard using Python and yfinance to track live Net Asset Value (NAV) feeds across equity, debt, and ETF asset allocations • **Developed** algorithmic financial features, including Buy/Hold/Exit signaling and an automated STCG/LTCG tax estimator, demonstrating practical application of capital gains taxation and personal finance strategy • **Optimized** Realized P&L and Yield to Maturity (YTM) calculations by resolving complex data conflicts across overlapping SIP cycles, ensuring high-fidelity financial reporting • **Integrated** an AI-driven market analyst module (Llama 3.3 70B) for macroeconomic insights, alongside automated daily performance alerts to monitor asset-liability alignment	2025
Travelogue — Team Trailblazer *Smart India Hackathon, Internal Round — Hack-O-Sprint 3.0, IIIT Bhopal | Project ID: 25137*	• **Led** a cross-functional team of developers as Team Leader to conceive and build "Travelogue," a tech-driven tourism application, from ideation through submission • **Cleared** a highly competitive internal selection round, securing the project's advancement to the official national-stage submission for Smart India Hackathon • **Managed** the full project lifecycle — task allocation, timeline planning, and mentor coordination — under competitive, deadline-driven conditions	2025

LEADERSHIP & EXTRACURRICULARS

Co-Coordinator *Shankh — College Magazine, IIIT Bhopal*	• **Co-coordinate** the editorial planning and publication of Shankh, the official college magazine of IIIT Bhopal, overseeing content curation, contributor outreach and publication deadlines • **Drive** editorial quality and contributor engagement across the student community, drawing on sustained long-form writing experience	2026–2027
Analytical Writing *Substack — "The Poetic Generalist"*	• **Author** a newsletter of original long-form essays spanning philosophical commentary, political history, and cultural analysis, built to an engaged readership of **1000+ subscribers and 1.5k+ followers** • Demonstrates sustained capability in independent research, structured argumentation, and long-form analytical writing outside the classroom	Ongoing
Portfolio Management *Personal Investing*	• **Manage** a diversified mutual fund SIP portfolio via Groww, allocated across domestic equity, international equity, gold, and arbitrage funds to optimize for non-correlated returns • Maintain active interest in algorithmic trading, data-driven economic modelling, and personal finance strategy	Ongoing

AWARDS & ACHIEVEMENTS

Public Speaking *Competitive Quizzing & Debate*	• **Selected** for Zee TV's national quiz program "Mind Wars" and secured 2nd Position at both the WWF Wide Wisdom Quiz (Regional – MP & CG) and the State Road Safety Debate, felicitated by the State Transport Minister	2019
Scholastic Excellence *Academic Olympiads*	• **Earned** the SOF International GK Olympiad's Certificate of Zonal Excellence and Medal of Distinction, backed by a consistent record as a top academic performer and multiple school-level Olympiad wins across subjects	2019–20

TECHNICAL SKILLS

Languages	C, C++17/20, Python 3; HTML5, CSS3, JavaScript (ES6+)
Frameworks & Tools	Flask, Streamlit, pandas, NumPy, Plotly, SQL, PostgreSQL, yfinance, pytest, Playwright, Git, GitHub, GitHub Actions, VS Code, XAMPP
APIs & AI	FRED API, Groq API (Llama 3.3 70B), Anthropic Claude API, python-dotenv
Finance	Mutual Fund Analysis, SIP Portfolio Management, STCG/LTCG Taxation, Market Research
FinTech & Quant	Quantitative Data Engineering, Algorithmic Backtesting, Derivatives & Options Pricing (Black-Scholes, Greeks, Monte Carlo), API Data Integration (FRED/yfinance), Financial Modeling, Loan & EMI Modelling
Concepts	Data Structures & Algorithms, OOP, DBMS, Computer Networks, Compiler Design